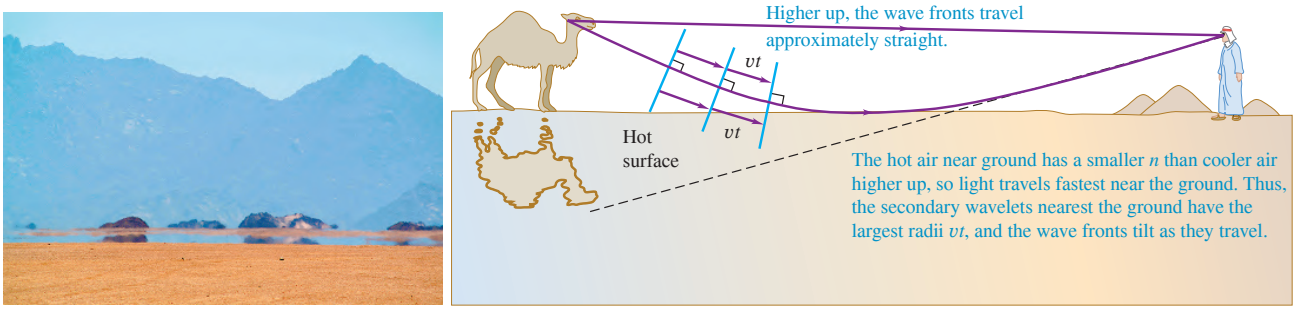
Figure **33.36** How mirages are formed.



TEST YOUR UNDERSTANDING OF SECTION 33.7 Sound travels faster in warm air than in cold air. Imagine a weather front that runs north-south, with warm air to the west of the front and cold air to the east. A sound wave traveling in a northeast direction in the warm air encounters this front. How will the direction of this sound wave change when it passes into the cold air? (i) The wave direction will deflect toward the north; (ii) the wave direction will deflect toward the east; (iii) the wave direction will be unchanged.

ANSWER

(ii) Huygens's principle applies to waves of all kinds, including sound waves. Hence this situation is exactly like that shown in Fig. 33.35, with material *a* representing the warm air, material *b* representing the cold air in which the waves travel more slowly, and the interface between the materials representing the weather front. North is toward the top of the figure and east is toward the right, so Fig. 33.35 shows that the rays (which indicate the direction of propagation) deflect toward the east.

CHAPTER 33 SUMMARY

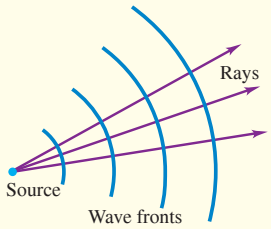
Light and its properties: Light is an electromagnetic wave. When emitted or absorbed, it also shows particle properties. It is emitted by accelerated electric charges.

A wave front is a surface of constant phase; wave fronts move with a speed equal to the propagation speed of the wave. A ray is a line along the direction of propagation, perpendicular to the wave fronts.

When light is transmitted from one material to another, the frequency of the light is unchanged, but the wavelength and wave speed can change. The index of refraction n of a material is the ratio of the speed of light in vacuum c to the speed v in the material. If λ_0 is the wavelength in vacuum, the same wave has a shorter wavelength λ in a medium with index of refraction n. (See Example 33.2.)

$$n = \frac{c}{v} \quad (33.1)$$

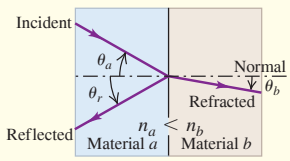
$$\lambda = \frac{\lambda_0}{n} \quad (33.5)$$



Reflection and refraction: At a smooth interface between two optical materials, the incident, reflected, and refracted rays and the normal to the interface all lie in a single plane called the plane of incidence. The law of reflection states that the angles of incidence and reflection are equal. The law of refraction relates the angles of incidence and refraction to the indexes of refraction of the materials. (See Examples 33.1 and 33.3.)

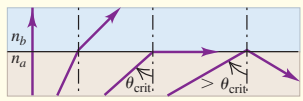
$$\theta_r = \theta_a \quad (33.2)$$
(law of reflection)

$$n_a \sin\theta_a = n_b \sin\theta_b \quad (33.4)$$
(law of refraction)



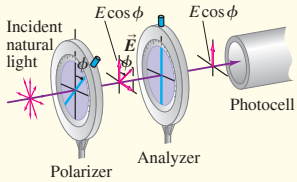
Total internal reflection: When a ray travels in a material of index of refraction n_a toward a material of index $n_b < n_a$, total internal reflection occurs at the interface when the angle of incidence equals or exceeds a critical angle θ_{crit}. (See Example 33.4.)

$$\sin\theta_{crit} = \frac{n_b}{n_a} \quad (33.6)$$



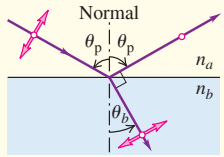
Polarization of light: The direction of polarization of a linearly polarized electromagnetic wave is the direction of the $\vec{E}$ field. A polarizing filter passes waves that are linearly polarized along its polarizing axis and blocks waves polarized perpendicularly to that axis. When polarized light of intensity I_{max} is incident on a polarizing filter used as an analyzer, the intensity I of the light transmitted through the analyzer depends on the angle ϕ between the polarization direction of the incident light and the polarizing axis of the analyzer. (See Example 33.5.)

$$I = I_{max}\cos^2\phi \quad \text{(Malus's law)} \qquad (33.7)$$

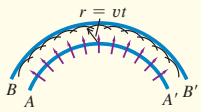


Polarization by reflection: When unpolarized light strikes an interface between two materials, Brewster's law states that the reflected light is completely polarized perpendicular to the plane of incidence (parallel to the interface) if the angle of incidence equals the polarizing angle θ_p. (See Example 33.6.)

$$\tan\theta_p = \frac{n_b}{n_a} \quad \text{(Brewster's law)} \qquad (33.8)$$



Huygens's principle: Huygens's principle states that if the position of a wave front at one instant is known, then the position of the front at a later time can be constructed by imagining the front as a source of secondary wavelets. Huygens's principle can be used to derive the laws of reflection and refraction.



Chapter 33 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 33.1 and 33.2 (Section 33.2) before attempting these problems.

VP33.2.1 A block of glass with index of refraction 1.80 has a smooth surface. Light in air strikes this surface at an angle of incidence of 70.0° measured from the normal to the surface of the glass. Find the angles measured relative to this normal of (a) the reflected ray and (b) the refracted ray.

VP33.2.2 A ray of light in water ($n = 1.33$) strikes a submerged glass block at an angle of incidence of 55.0°. The angle of refraction for the light that enters the glass is 37.0°. Find (a) the index of refraction of the glass and (b) the speed of light in the glass.

VP33.2.3 The light from a red laser pointer has wavelength 635 nm in air and 508 nm in a transparent liquid. You point the laser in air so that the beam strikes the surface of the liquid at an angle of 35.0° from the normal. Find (a) the index of refraction of the liquid, (b) the angle of refraction, (c) the frequency of the light in air, and (d) the frequency of the light in the liquid.

VP33.2.4 A glass of ethanol ($n = 1.36$) has an ice cube ($n = 1.309$) floating in it. A light beam in the ethanol goes into the ice cube at an angle of refraction of 85.0°. Find (a) the angle of incidence in the ethanol and (b) the ratio of the wavelength of the light in ice to its wavelength in ethanol.

Be sure to review EXAMPLE 33.5 (Section 33.5) before attempting these problems.

VP33.5.1 A polarized laser beam of intensity 255 W/m^2 shines on an ideal polarizer. The angle between the polarization direction of the laser beam and the polarizing axis of the polarizer is 15.0°. What is the intensity of the light that emerges from the polarizer?

VP33.5.2 You shine unpolarized light with intensity 54.0 W/m^2 on an ideal polarizer, and then the light that emerges from this polarizer falls on a second ideal polarizer. The light that emerges from the second polarizer has intensity 19.0 W/m^2. Find (a) the intensity of the light that emerges from the first polarizer and (b) the angle between the polarizing axes of the two polarizers.

VP33.5.3 A beam of polarized light of intensity 60.0 W/m^2 propagates in the $+x$-direction. The light is polarized in the $+y$-direction. The beam strikes an ideal polarizer whose plane is parallel to the yz-plane and has its polarizing axis at 25.0° clockwise from the y-direction. Then the beam that emerges from this polarizer strikes a second ideal polarizer whose plane is also parallel to the yz-plane but has its polarizing axis at 50.0° clockwise from the y-direction. Find the intensity of the light that emerges (a) from the first polarizer, (b) from the second polarizer, and (b) from the second polarizer if the first polarizer is removed.

VP33.5.4 You simultaneously shine two light beams, each of intensity I_0, on an ideal polarizer. One beam is unpolarized, and the other beam is polarized at an angle of exactly 30° to the polarizing axis of the polarizer. Find the intensity of the light that emerges from the polarizer.

Be sure to review EXAMPLE 33.6 (Section 33.5) before attempting these problems.

VP33.6.1 Unpolarized sunlight in air shines on a block of a transparent solid with index of refraction 1.73. (a) For what angle of incidence is the reflected light completely polarized? (b) For this angle of incidence, is the light refracted into the solid completely polarized, partially polarized, or unpolarized?

VP33.6.2 You shine a beam of unpolarized light in air on a block of glass. You find that if the angle of incidence is 57.0°, the reflected light is completely polarized. Find (a) the index of refraction of the glass and (b) the angle of refraction.

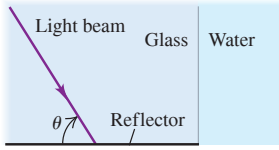
VP33.6.3 You shine a beam of *polarized* light in air on a piece of dense flint glass ($n = 1.66$). (a) If the polarization direction is perpendicular to the plane of incidence, is there an angle of incidence for which no light is reflected from the glass? If so, what is this angle? (b) Repeat part (a) if the polarization direction is in the plane of incidence.

VP33.6.4 A glass container holds water ($n = 1.33$). If unpolarized light propagating in the glass strikes the glass–water interface, the light reflected back into the glass will be completely polarized if the angle of refraction is 53.5°. Find (a) the polarizing angle in this situation and (b) the index of refraction of the glass.

BRIDGING PROBLEM Reflection and Refraction

Figure 33.37 shows a rectangular glass block that has a metal reflector on one face and water on an adjoining face. A light beam strikes the reflector as shown. You gradually increase the angle θ of the light beam. If $\theta \geq 59.2°$, no light enters the water. What is the speed of light in this glass?

Figure 33.37 Glass bounded by water and a metal reflector.



SOLUTION GUIDE

IDENTIFY and SET UP

1. Specular reflection occurs where the light ray in the glass strikes the reflector. If no light is to enter the water, we require that there be reflection only and no refraction where this ray strikes the glass–water interface—that is, there must be total internal reflection.
2. The target variable is the speed of light v in the glass, which you can determine from the index of refraction n of the glass. (Table 33.1 gives the index of refraction of water.) Write down the equations you'll use to find n and v.

EXECUTE

3. Use the figure to find the angle of incidence of the ray at the glass–water interface.
4. Use the result of step 3 to find n.
5. Use the result of step 4 to find v.

EVALUATE

6. How does the speed of light in the glass compare to the speed in water? Does this make sense?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q33.1 Light requires about 8 minutes to travel from the sun to the earth. Is it delayed appreciably by the earth's atmosphere? Explain.

Q33.2 Sunlight or starlight passing through the earth's atmosphere is always bent toward the vertical. Why? Does this mean that a star is not really where it appears to be? Explain.

Q33.3 A beam of light goes from one material into another. On *physical* grounds, explain *why* the wavelength changes but the frequency and period do not.

Q33.4 A student claimed that, because of atmospheric refraction (see Discussion Question Q33.2), the sun can be seen after it has set and that the day is therefore longer than it would be if the earth had no atmosphere. First, what does she mean by saying that the sun can be seen after it has set? Second, comment on the validity of her conclusion.

Q33.5 When hot air rises from a radiator or heating duct, objects behind it appear to shimmer or waver. What causes this?

Q33.6 Devise straightforward experiments to measure the speed of light in a given glass using (a) Snell's law; (b) total internal reflection; (c) Brewster's law.

Q33.7 Sometimes when looking at a window, you see two reflected images slightly displaced from each other. What causes this?

Q33.8 If you look up from underneath toward the surface of the water in your aquarium, you may see an upside-down reflection of your pet fish in the surface of the water. Explain how this can happen.

Q33.9 A ray of light in air strikes a glass surface. Is there a range of angles for which total internal reflection occurs? Explain.

Q33.10 When light is incident on an interface between two materials, the angle of the refracted ray depends on the wavelength, but the angle of the reflected ray does not. Why should this be?

Q33.11 A salesperson at a bargain counter claims that a certain pair of sunglasses has Polaroid filters; you suspect that the glasses are just tinted plastic. How could you find out for sure?

Q33.12 Does it make sense to talk about the polarization of a *longitudinal* wave, such as a sound wave? Why or why not?

Q33.13 How can you determine the direction of the polarizing axis of a single polarizer?

Q33.14 It has been proposed that automobile windshields and headlights should have polarizing filters to reduce the glare of oncoming lights during night driving. Would this work? How should the polarizing axes be arranged? What advantages would this scheme have? What disadvantages?

Q33.15 When a sheet of plastic food wrap is placed between two crossed polarizers, no light is transmitted. When the sheet is stretched in one direction, some light passes through the crossed polarizers. What is happening?

Q33.16 If you sit on the beach and look at the ocean through Polaroid sunglasses, the glasses help to reduce the glare from sunlight reflecting off the water. But if you lie on your side on the beach, there is little reduction in the glare. Explain why there is a difference.

Q33.17 When unpolarized light is incident on two crossed polarizers, no light is transmitted. A student asserted that if a third polarizer is inserted between the other two, some transmission will occur. Does this make sense? How can adding a third filter *increase* transmission?

Q33.18 For the old "rabbit-ear" style TV antennas, it's possible to alter the quality of reception considerably simply by changing the orientation of the antenna. Why?

Q33.19 In Fig. 33.31, since the light that is scattered out of the incident beam is polarized, why is the transmitted beam not also partially polarized?

Q33.20 You are sunbathing in the late afternoon when the sun is relatively low in the western sky. You are lying flat on your back, looking straight up through Polaroid sunglasses. To minimize the amount of sky light reaching your eyes, how should you lie: with your feet pointing north, east, south, west, or in some other direction? Explain your reasoning.

Q33.21 Light scattered from blue sky is strongly polarized because of the nature of the scattering process described in Section 33.6. But light scattered from white clouds is usually *not* polarized. Why not?

Q33.22 Atmospheric haze is due to water droplets or smoke particles ("smog"). Such haze reduces visibility by scattering light, so that the light from distant objects becomes randomized and images become indistinct. Explain why visibility through haze can be improved by wearing red-tinted sunglasses, which filter out blue light.

Q33.23 The explanation given in Section 33.6 for the color of the setting sun should apply equally well to the *rising* sun, since sunlight travels the same distance through the atmosphere to reach your eyes at either sunrise or sunset. Typically, however, sunsets are redder than sunrises. Why? (*Hint:* Particles of all kinds in the atmosphere contribute to scattering.)

Q33.24 Huygens's principle also applies to sound waves. During the day, the temperature of the atmosphere decreases with increasing altitude above the ground. But at night, when the ground cools, there is a layer of air just above the surface in which the temperature *increases* with altitude. Use this to explain why sound waves from distant sources can be heard more clearly at night than in the daytime. (*Hint:* The speed of sound increases with increasing temperature. Use the ideas displayed in Fig. 33.36 for light.)

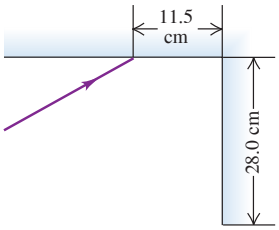
Q33.25 Can water waves be reflected and refracted? Give examples. Does Huygens's principle apply to water waves? Explain.

EXERCISES

Section 33.2 Reflection and Refraction

33.1 • Two plane mirrors intersect at right angles. A laser beam strikes the first of them at a point 11.5 cm from their point of intersection, as shown in **Fig. E33.1**. For what angle of incidence at the first mirror will this ray strike the midpoint of the second mirror (which is 28.0 cm long) after reflecting from the first mirror?

Figure E33.1



33.2 • **BIO Light Inside the Eye.** The vitreous humor, a transparent, gelatinous fluid that fills most of the eyeball, has an index of refraction of 1.34. Visible light ranges in wavelength from 380 nm (violet) to 750 nm (red), as measured in air. This light travels through the vitreous humor and strikes the rods and cones at the surface of the retina. What are the ranges of (a) the wavelength, (b) the frequency, and (c) the speed of the light just as it approaches the retina within the vitreous humor?

33.3 • A beam of light has a wavelength of 620 nm in vacuum. (a) What is the speed of this light in a liquid whose index of refraction at this wavelength is 1.42 (b) What is the wavelength of these waves in the liquid?

33.4 • Light with a frequency of 5.45×10^{14} Hz travels in a block of glass that has an index of refraction of 1.55. What is the wavelength of the light (a) in vacuum and (b) in the glass?

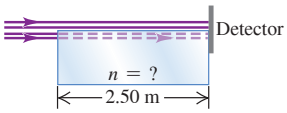
33.5 • A liquid lies on top of the horizontal surface of a block of glass. A ray of light traveling in the glass has speed 1.85×10^8 m/s, wavelength 365 nm, and frequency 5.07×10^{14} Hz. The ray is incident on the surface of the glass at an angle of 38.0° with respect to the normal to the surface. The ray that refracts into the liquid makes an angle of 44.7° with the normal to the interface between the two materials. What are the speed, wavelength, and frequency of the light when it is traveling in the liquid?

33.6 •• Light of a certain frequency has a wavelength of 526 nm in water. What is the wavelength of this light in benzene?

33.7 •• A parallel beam of light in air makes an angle of 41.0° with the surface of a glass plate having a refractive index of 1.57. (a) What is the angle between the reflected part of the beam and the surface of the glass? (b) What is the angle between the refracted beam and the surface of the glass?

33.8 •• A laser beam shines along the surface of a block of transparent material (see **Fig. E33.8**). Half of the beam goes straight to a detector, while the other half travels through the block and then hits the detector. The time delay between the arrival of the two light beams at the detector is 6.15 ns. What is the index of refraction of this material?

Figure E33.8

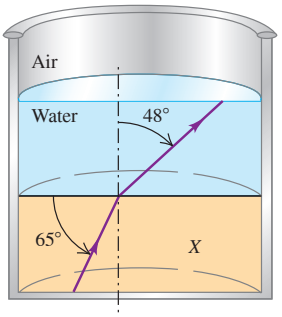


33.9 • Light traveling in air is incident on the surface of a block of plastic at an angle of 60.7° to the normal and is bent so that it makes a 45.5° angle with the normal in the plastic. Find the speed of light in the plastic.

33.10 • (a) A tank containing methanol has walls 2.50 cm thick made of glass of refractive index 1.550. Light from the outside air strikes the glass at a 41.3° angle with the normal to the glass. Find the angle the light makes with the normal in the methanol. (b) The tank is emptied and refilled with an unknown liquid. If light incident at the same angle as in part (a) enters the liquid in the tank at an angle of 20.2° from the normal, what is the refractive index of the unknown liquid?

33.11 •• As shown in **Fig. E33.11**, a layer of water covers a slab of material *X* in a beaker. A ray of light traveling upward follows the path indicated. Using the information on the figure, find (a) the index of refraction of material *X* and (b) the angle the light makes with the normal in the *air*.

Figure E33.11



33.12 •• A horizontal, parallel-sided plate of glass having a refractive index of 1.50 is in contact with the surface of water in a tank. A ray coming from above in air makes an angle of incidence of 33.0° with the normal to the top surface of the glass. (a) What angle does the ray refracted into the water make with the normal to the surface? (b) What is the dependence of this angle on the refractive index of the glass?

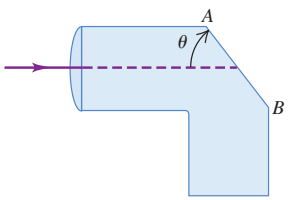
33.13 • A block of a transparent solid sits on top of the horizontal surface of a block of glass. A ray of light traveling in the glass is incident on the top surface of the glass at an angle of 62.0° with respect to the normal to the surface. The light has wavelength 447 nm in the glass and 315 nm in the transparent solid. What angle does the ray that refracts into the transparent solid make with the normal to the surface?

33.14 • A ray of light traveling in water is incident on an interface with a flat piece of glass. The wavelength of the light in the water is 727 nm, and its wavelength in the glass is 542 nm. If the ray in water makes an angle of 41.5° with respect to the normal to the interface, what angle does the refracted ray in the glass make with respect to the normal?

Section 33.3 Total Internal Reflection

33.15 • **Light Pipe.** Light enters a solid pipe made of plastic having an index of refraction of 1.60. The light travels parallel to the upper part of the pipe (**Fig. E33.15**). You want to cut the face *AB* so that all the light will reflect back into the pipe after it first strikes that face. (a) What is the largest that θ can be if the pipe is in air? (b) If the pipe is immersed in water of refractive index 1.33, what is the largest that θ can be?

Figure E33.15



33.16 • A flat piece of glass covers the top of a vertical cylinder that is completely filled with water. If a ray of light traveling in the glass is incident on the interface with the water at an angle of $\theta_a = 36.2°$, the ray refracted into the water makes an angle of 49.8° with the normal to the interface. What is the smallest value of the incident angle θ_a for which none of the ray refracts into the water?

33.17 •• The critical angle for total internal reflection at a liquid–air interface is 44.5°. (a) If a ray of light traveling in the liquid has an angle of incidence at the interface of 32.5°, what angle does the refracted ray in the air make with the normal? (b) If a ray of light traveling in air has an angle of incidence at the interface of 32.5°, what angle does the refracted ray in the liquid make with the normal?

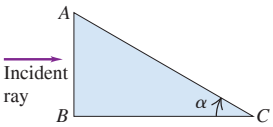
33.18 • A beam of light is traveling inside a solid glass cube that has index of refraction 1.53. It strikes the surface of the cube from the inside. (a) If the cube is in air, at what minimum angle with the normal inside the glass will this light *not* enter the air at this surface? (b) What would be the minimum angle in part (a) if the cube were immersed in water?

33.19 • A ray of light is traveling in a glass cube that is totally immersed in water. You find that if the ray is incident on the glass–water interface at an angle to the normal larger than 46.0°, no light is refracted into the water. What is the refractive index of the glass?

33.20 • At the very end of Wagner's series of operas *Ring of the Nibelung,* Brünnhilde takes the golden ring from the finger of the dead Siegfried and throws it into the Rhine, where it sinks to the bottom of the river. Assuming that the ring is small enough compared to the depth of the river to be treated as a point and that the Rhine is 13.9 m deep where the ring goes in, what is the area of the largest circle at the surface of the water over which light from the ring could escape from the water?

33.21 • Light is incident along the normal on face AB of a glass prism of refractive index 1.52, as shown in **Fig. E33.21**. Find the largest value the angle α can have without any light refracted out of the prism at face AC if (a) the prism is immersed in air and (b) the prism is immersed in water.

Figure **E33.21**

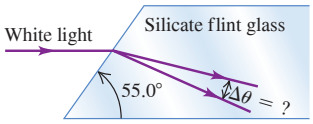


33.22 • A thick sheet of transparent plastic sits on top of the horizontal surface of a block of glass. For a ray traveling in the glass, the critical angle at the glass–plastic interface is 48.6° and the wavelength of the light in the glass is 350 nm. For an incident angle of 40.0° in the glass, what is the wavelength of the light after it refracts into the plastic?

Section 33.4 Dispersion

33.23 •• A narrow beam of white light strikes one face of a slab of silicate flint glass. The light is traveling parallel to the two adjoining faces, as shown in **Fig. E33.23**. For the transmitted light inside the glass, through what angle $\Delta\theta$ is the portion of the visible spectrum between 400 nm and 700 nm dispersed? (Consult the graph in Fig. 33.17.)

Figure **E33.23**



33.24 • The indexes of refraction for violet light ($\lambda = 400$ nm) and red light ($\lambda = 700$ nm) in diamond are 2.46 and 2.41, respectively. A ray of light traveling through air strikes the diamond surface at an angle of 53.5° to the normal. Calculate the angular separation between these two colors of light in the refracted ray.

33.25 • A beam of light strikes a sheet of glass at an angle of 57.0° with the normal in air. You observe that red light makes an angle of 38.1° with the normal in the glass, while violet light makes a 36.7° angle. (a) What are the indexes of refraction of this glass for these colors of light? (b) What are the speeds of red and violet light in the glass?

Section 33.5 Polarization

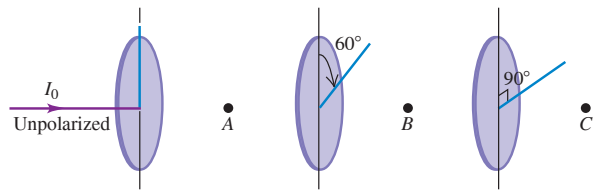
33.26 •• Birefringence is discussed in Section 33.5 and the refractive indexes for the two perpendicular polarization directions in calcite are given. A crystal of calcite serves as a quarter-wave plate; it converts linearly polarized light to circularly polarized light if the numbers of wavelengths within the crystal differ by one-fourth for the two polarization components. For light with wavelength 589 nm in air, what is the minimum thickness of a quarter-wave plate made of calcite?

33.27 • Unpolarized light with intensity I_0 is incident on two polarizing filters. The axis of the first filter makes an angle of 60.0° with the vertical, and the axis of the second filter is horizontal. What is the intensity of the light after it has passed through the second filter?

33.28 • A layer of liquid sits on top of the horizontal surface of a transparent solid. For a ray traveling in the solid and incident on the interface of the two materials, the critical angle is 38.7°. (a) For a ray traveling in the solid and reflecting at the interface with the liquid, for what incident angle with respect to the normal is the reflected ray 100% polarized? (b) What is the polarizing angle if the ray is traveling in the liquid?

33.29 •• A beam of unpolarized light of intensity I_0 passes through a series of ideal polarizing filters with their polarizing axes turned to various angles as shown in **Fig. E33.29**. (a) What is the light intensity (in terms of I_0) at points A, B, and C? (b) If we remove the middle filter, what will be the light intensity at point C?

Figure **E33.29**



33.30 •• Light of original intensity I_0 passes through two ideal polarizing filters having their polarizing axes oriented as shown in **Fig. E33.30**. You want to adjust the angle ϕ so that the intensity at point P is equal to $I_0/18$. (a) If the original light is unpolarized, what should ϕ be? (b) If the original light is linearly polarized in the same direction as the polarizing axis of the first polarizer the light reaches, what should ϕ be?

Figure **E33.30**

33.31 • A parallel beam of unpolarized light in air is incident at an angle of 54.0° (with respect to the normal) on a plane glass surface. The reflected beam is completely linearly polarized. (a) What is the refractive index of the glass? (b) What is the angle of refraction of the transmitted beam?

33.32 • The refractive index of a certain glass is 1.69. For what incident angle is light reflected from the surface of this glass completely polarized if the glass is immersed in (a) air and (b) water?

33.33 •• Unpolarized light of intensity 23.2 W/cm^2 is incident on two polarizing filters. The axis of the first filter is at an angle of 23.8° counterclockwise from the vertical (viewed in the direction the light is traveling), and the axis of the second filter is at 59.0° counterclockwise from the vertical. What is the intensity of the light after it has passed through the second polarizer?

33.34 ••• Three polarizing filters are stacked, with the polarizing axis of the second and third filters at 23.0° and 62.0°, respectively, clockwise to that of the first. If unpolarized light is incident on the stack, the light has intensity 55.0 W/cm^2 after it passes through the stack. If the incident intensity is kept constant but the second polarizer is removed, what is the intensity of the light after it has passed through the stack?

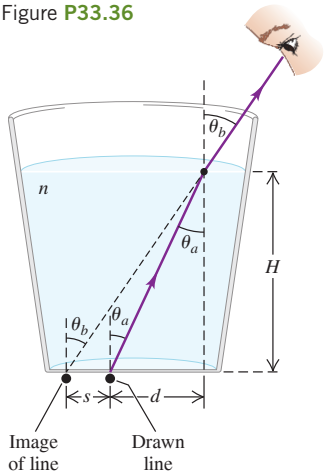
Section 33.6 Scattering of Light

33.35 • A beam of white light passes through a uniform thickness of air. If the intensity of the scattered light in the middle of the green part of the visible spectrum is I, find the intensity (in terms of I) of scattered light in the middle of (a) the red part of the spectrum and (b) the violet part of the spectrum. Consult Table 32.1.

PROBLEMS

33.36 •• Draw a straight line on a piece of paper. Fill a transparent drinking glass with water and place it on top of the line you have drawn, making sure the line extends outward from both sides of the glass. Position your head above the glass, but not directly above, such that you look downward at an angle to see the image of the line through the water, as shown in **Fig. P33.36**. The image of the line beneath the glass will appear shifted away from the line you have drawn. (a) Mark a spot on the paper beside the glass where the line seen through the water would extend. Estimate the distance s between that mark and the original line. (b) Keeping your eye in the same position, hold a pen along the line of sight from your eye to the surface of the water, where the image of the line appears. The tip of the pen should just touch the surface of the water. Now move your head above the glass and sight downward from directly above the tip of the pen. Mark a spot on the paper beside the glass indicating the horizontal distance between the drawn line and the tip of the pen. Estimate that distance d. (c) Let θ_a and θ_b represent the angle of incidence and the angle of refraction, respectively, of the upward light ray traveling toward your eye, as shown in the figure, and let H be the depth of the water. Write expressions for $\tan\theta_a$ and $\tan\theta_b$ in terms of s, d, and H. (d) Assume H is large so the angles are small and we can approximate $\tan\theta_a \approx \sin\theta_a$ and $\tan\theta_b \approx \sin\theta_b$. Using Snell's law and these approximations, write an expression for the index of refraction n of the water in terms of s and d. (e) Use your values of s and d to estimate the index of refraction of the water.

Figure P33.36

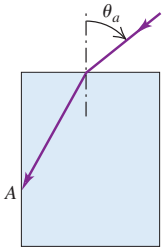


33.37 •• BIO **Heart Sonogram.** Physicians use high-frequency (f = 1–5 MHz) sound waves, called ultrasound, to image internal organs. The speed of these ultrasound waves is 1480 m/s in muscle and 344 m/s in air. We define the index of refraction of a material for sound waves to be the ratio of the speed of sound in air to the speed of sound in the material. Snell's law then applies to the refraction of sound waves. (a) At what angle from the normal does an ultrasound beam enter the heart if it leaves the lungs at an angle of 9.73° from the normal to the heart wall? (Assume that the speed of sound in the lungs is 344 m/s.) (b) What is the critical angle for sound waves in air incident on muscle?

33.38 ••• In a physics lab, light with wavelength 530 nm travels in air from a laser to a photocell in 16.7 ns. When a slab of glass 0.850 m thick is placed in the light beam, with the beam incident along the normal to the parallel faces of the slab, it takes the light 20.8 ns to travel from the laser to the photocell. What is the wavelength of the light in the glass?

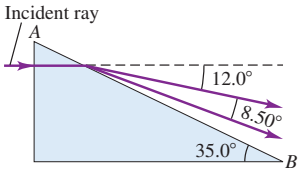
33.39 •• A ray of light is incident in air on a block of a transparent solid whose index of refraction is n. If n = 1.37, what is the *largest* angle of incidence θ_a for which total internal reflection will occur at the vertical face (point A shown in **Fig. P33.39**)?

Figure P33.39



33.40 • A light ray in air strikes the right-angle prism shown in **Fig. P33.40**. The prism angle at B is 35.0°. This ray consists of two different wavelengths. When it emerges at face AB, it has been split into two different rays that diverge from each other by 8.50°. Find the index of refraction of the prism for each of the two wavelengths.

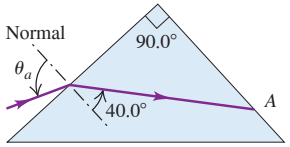
Figure P33.40



33.41 •• A ray of light traveling *in* a block of glass (n = 1.52) is incident on the top surface at an angle of 57.2° with respect to the normal in the glass. If a layer of oil is placed on the top surface of the glass, the ray is totally reflected. What is the maximum possible index of refraction of the oil?

33.42 •• A ray of light traveling in air is incident at angle θ_a on one face of a 90.0° prism made of glass. Part of the light refracts into the prism and strikes the opposite face at point A (**Fig. P33.42**). If the ray at A is at the critical angle, what is the value of θ_a?

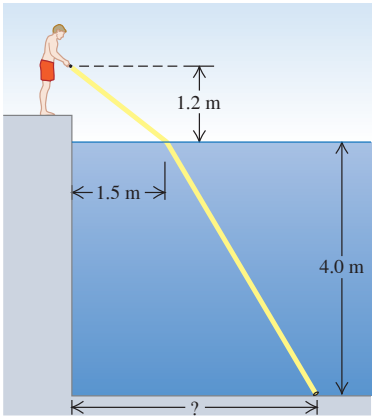
Figure P33.42



33.43 ••• A glass plate 2.65 mm thick, with an index of refraction of 1.80, is placed between a point source of light with wavelength 600 nm (in vacuum) and a screen. The distance from source to screen is 1.30 cm. How many wavelengths are there between the source and the screen?

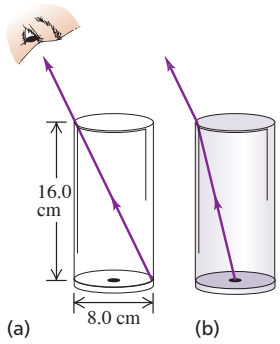
33.44 • After a long day of driving you take a late-night swim in a motel swimming pool. When you go to your room, you realize that you have lost your room key in the pool. You borrow a powerful flashlight and walk around the pool, shining the light into it. The light shines on the key, which is lying on the bottom of the pool, when the flashlight is held 1.2 m above the water surface and is directed at the surface a horizontal distance of 1.5 m from the edge (**Fig. P33.44**). If the water here is 4.0 m deep, how far is the key from the edge of the pool?

Figure P33.44



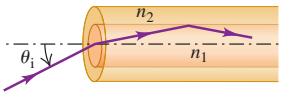
33.45 • You sight along the rim of a glass with vertical sides so that the top rim is lined up with the opposite edge of the bottom (**Fig. P33.45a**). The glass is a thin-walled, hollow cylinder 16.0 cm high. The diameter of the top and bottom of the glass is 8.0 cm. While you keep your eye in the same position, a friend fills the glass with a transparent liquid, and you then see a dime that is lying at the center of the bottom of the glass (Fig. P33.45b). What is the index of refraction of the liquid?

Figure P33.45



33.46 •• Optical fibers are constructed with a cylindrical core surrounded by a sheath of cladding material. Common materials used are pure silica ($n_2 = 1.450$) for the cladding and silica doped with germanium ($n_1 = 1.465$) for the core. (a) What is the critical angle θ_{crit} for light traveling in the core and reflecting at the interface with the cladding material? (b) The numerical aperture (NA) is defined as the angle of incidence θ_i at the flat end of the cable for which light is incident on the core–cladding interface at angle θ_{crit} (**Fig. P33.46**). Show that $\sin\theta_i = \sqrt{n_1^2 - n_2^2}$. (c) What is the value of θ_i for $n_1 = 1.465$ and $n_2 = 1.450$?

Figure P33.46

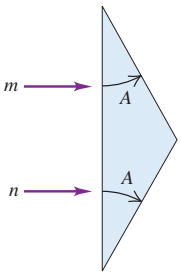


33.47 • A thin layer of ice ($n = 1.309$) floats on the surface of water ($n = 1.333$) in a bucket. A ray of light from the bottom of the bucket travels upward through the water. (a) What is the largest angle with respect to the normal that the ray can make at the ice–water interface and still pass out into the air above the ice? (b) What is this angle after the ice melts?

33.48 •• When linearly polarized light passes through a polarizer, its polarizing axis may be rotated by any angle $\phi < 90°$ at the expense of a loss of intensity, as determined by Malus's law. By using sequential polarizers, you can achieve a similar axis rotation but retain greater intensity. In fact, if you use many intermediate polarizers, the polarization axis can be rotated by 90° with virtually undiminished intensity. (a) Derive an equation for the resulting intensity if linearly polarized light passes through successive N polarizers, each with the polarizing axis rotated by an angle $90°/2N$ larger than the preceding polarizer. (b) By making a table of the resulting intensity for various values of N, estimate the minimum number N of polarizers needed so that the light will have its polarization axis rotated by 90° while maintaining more than 90% of its intensity. (c) Estimate the minimum number of polarizers needed to maintain more than 95% and 99% intensity.

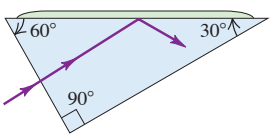
33.49 •• The prism shown in **Fig. P33.49** has a refractive index of 1.56, and the angles A are 27.0°. Two light rays m and n are parallel as they enter the prism. What is the angle between them after they emerge?

Figure P33.49



33.50 •• Light is incident normally on the short face of a 30°–60°–90° prism (**Fig. P33.50**). A drop of liquid is placed on the hypotenuse of the prism. If the index of refraction of the prism is 1.65, find the maximum index that the liquid may have for the light to be totally reflected.

Figure P33.50

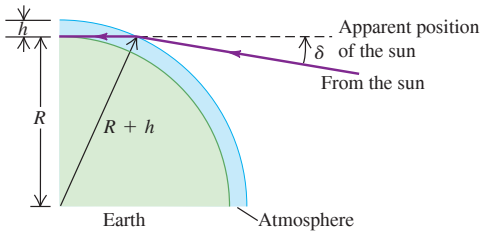


33.51 •• When the sun is either rising or setting and appears to be just on the horizon, it is in fact *below* the horizon. The explanation for this seeming paradox is that light from the sun bends slightly when entering the earth's atmosphere, as shown in **Fig. P33.51**. Since our perception is based on the idea that light travels in straight lines, we perceive the light to be coming from an apparent position that is an angle δ above the sun's true position. (a) Make the simplifying assumptions that the atmosphere has uniform density, and hence uniform index of refraction n, and extends to a height h above the earth's surface, at which point it abruptly stops. Show that the angle δ is given by

$$\delta = \arcsin\left(\frac{nR}{R+h}\right) - \arcsin\left(\frac{R}{R+h}\right)$$

where $R = 6378$ km is the radius of the earth. (b) Calculate δ using $n = 1.0003$ and $h = 20$ km. How does this compare to the angular radius of the sun, which is about one quarter of a degree? (In actuality a light ray from the sun bends gradually, not abruptly, since the density and refractive index of the atmosphere change gradually with altitude.)

Figure P33.51

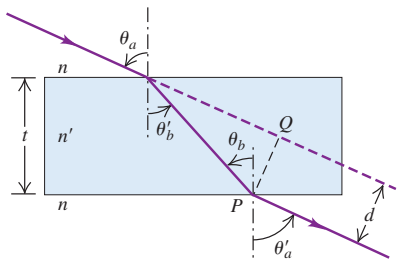


33.52 •• Light is incident in air at an angle θ_a (**Fig. P33.52**) on the upper surface of a transparent plate, the surfaces of the plate being plane and parallel to each other. (a) Prove that $\theta_a = \theta'_a$. (b) Show that this is true for any number of different parallel plates. (c) Prove that the lateral displacement d of the emergent beam is given by the relationship

$$d = t\frac{\sin(\theta_a - \theta'_b)}{\cos\theta'}$$

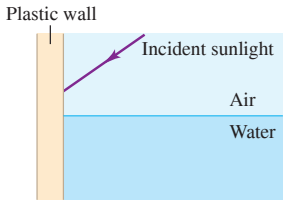
where t is the thickness of the plate. (d) A ray of light is incident at an angle of 66.0° on one surface of a glass plate 2.40 cm thick with an index of refraction of 1.80. The medium on either side of the plate is air. Find the lateral displacement between the incident and emergent rays.

Figure P33.52



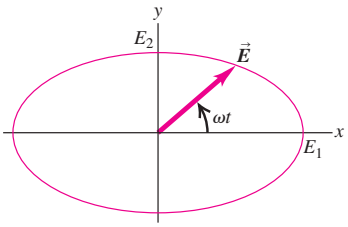
33.53 •• A beam of unpolarized sunlight strikes the vertical plastic wall of a water tank at an unknown angle. Some of the light reflects from the wall and enters the water (**Fig. P33.53**). The refractive index of the plastic wall is 1.67. If the light that has been reflected from the wall into the water is observed to be completely polarized, what angle does this beam make with the normal inside the water?

Figure P33.53



33.54 ••• **CP** A circularly polarized electromagnetic wave propagating in air has an electric field given by $\vec{E} = E[\cos(kz - \omega t)\,\hat{\imath} + \sin(kz - \omega t)\,\hat{\jmath}]$. This wave is incident with an intensity of 150 W/m^2 at the polarizing angle θ_p onto a flat interface perpendicular to the xz-plane with a material that has index of refraction $n = 1.62$. (a) What is the angle of refraction θ_b? (b) The ratio between the electric-field amplitude in the reflected wave and the electric-field amplitude of the component of the incident wave polarized parallel to the interface is $\sin(\theta_p - \theta_b)/\sin(\theta_p + \theta_b)$. (This result, known as the Fresnel equation, may be derived using deeper analysis.) From this relationship, determine the intensity $I_{\text{reflected}}$ of the reflected wave. (c) Determine the intensity $I_{\parallel}$ of the component of the refracted wave polarized parallel to the interface. (d) Determine the intensity $I_{\perp}$ of the component of the refracted wave polarized in the x-direction. (e) The reflected wave is linearly polarized and the refracted wave is elliptically polarized, such that its electric field is characterized as shown in **Fig. P33.54**. Determine the elliptical eccentricity $e = \sqrt{1 - (E_1/E_2)^2}$.

Figure **P33.54**



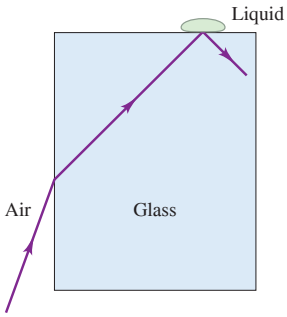
33.55 •• **DATA** In physics lab, you are studying the properties of four transparent liquids. You shine a ray of light (in air) onto the surface of each liquid—A, B, C, and D—one at a time, at a 60.0° angle of incidence; you then measure the angle of refraction. The table gives your data:

Liquid	A	B	C	D
θ_a (°)	36.4	40.5	32.1	35.2

The wavelength of the light when it is traveling in air is 589 nm. (a) Find the refractive index of each liquid at this wavelength. Use Table 33.1 to identify each liquid, assuming that all four are listed in the table. (b) For each liquid, what is the dielectric constant K at the frequency of the 589 nm light? For each liquid, the relative permeability (K_m) is very close to unity. (c) What is the frequency of the light in air and in each liquid?

33.56 •• **DATA** Given small samples of three liquids, you are asked to determine their refractive indexes. However, you do not have enough of each liquid to measure the angle of refraction for light refracting from air into the liquid. Instead, for each liquid, you take a rectangular block of glass ($n = 1.52$) and place a drop of the liquid on the top surface of the block. You shine a laser beam with wavelength 638 nm in vacuum at one side of the block and measure the largest angle of incidence θ_a for which there is total internal reflection at the interface between the glass and the liquid (**Fig. P33.56**). Your results are given in the table:

Figure **P33.56**

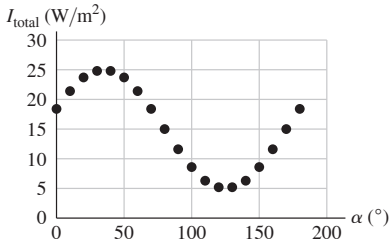


Liquid	A	B	C
θ_a (°)	52.0	44.3	36.3

What is the refractive index of each liquid at this wavelength?

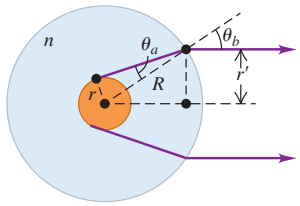
33.57 •• **DATA** A beam of light traveling horizontally is made of an unpolarized component with intensity I_0 and a polarized component with intensity I_p. The plane of polarization of the polarized component is oriented at an angle θ with respect to the vertical. **Figure P33.57** is a graph of the total intensity I_{total} after the light passes through a polarizer versus the angle α that the polarizer's axis makes with respect to the vertical. (a) What is the orientation of the polarized component? (That is, what is θ?) (b) What are the values of I_0 and I_p?

Figure **P33.57**



33.58 •• **CP** An object surrounded by a translucent material with index of refraction greater than unity appears enlarged. This includes items such as plants or shells inside plastic or resin paperweights, as well as ancient bugs trapped in amber. Consider a spherical object in air with radius r fixed at the center of a sphere with radius $R > r$ with refractive index n, as shown in **Fig. P33.58**. When viewed from outside, the spherical object appears to have radius $r' > r$. (a) Using right triangles found in the figure, write an expression for $\sin\theta_a$ in terms of r and R. (b) Similarly, write an expression for $\sin\theta_b$ in terms of r' and R. (c) Using Snell's law, derive an expression for r' in terms of r, R, and n. (d) If the object is a spherical dandelion seed head with diameter 45.0 mm fixed at the center of a solidified resin sphere with radius 80.0 mm and index of refraction 1.53, what diameter will the seed head appear to have when viewed from outside?

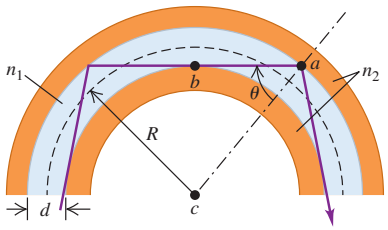
Figure **P33.58**



CHALLENGE PROBLEMS

33.59 ••• A fiber-optic cable consists of a thin cylindrical core with thickness d made of a material with index of refraction n_1, surrounded by cladding made of a material with index of refraction $n_2 < n_1$. Light rays traveling within the core remain trapped in the core provided they do not strike the core–cladding interface at an angle larger than the critical angle for total internal reflection. (a) **Figure P33.59** shows a light ray traveling within a cable bent to its critical extent, beyond which light will leak out of the core. The radius of curvature of the center of the core is R. Derive an expression for the angle of incidence θ at which the indicated light ray strikes the outer edge of the core. (*Hint:* Construct a right triangle with vertices a, b, and c as in the figure.) (b) Determine the

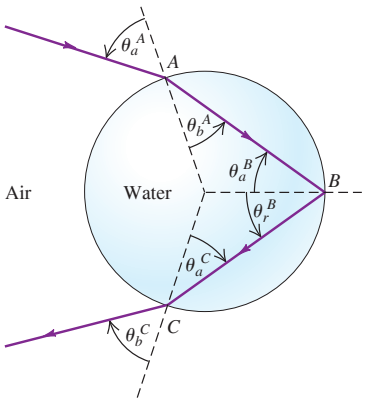
Figure **P33.59**



critical radius R in terms of n_1, n_2, and d. (c) A typical fiber-optic cable has a 50.0-μm-thick core with index of refraction 1.4475 and has a cladding with index of refraction 1.4440. What is the minimum radius of curvature for this cable? (d) If a light ray traversed a 1.00 km length of such a cable bent to its critical extent, and if the ray were maximally angled within the core as shown in the figure, what would be the extra distance, beyond 1.00 km, traveled by the ray owing to its multiple reflections? Keep extra digits in intermediate steps to reduce rounding errors. (e) How much longer would it take light to travel 1.00 km through the cable than 1.00 km in air?

33.60 ••• **CALC** A rainbow is produced by the reflection of sunlight by spherical drops of water in the air. **Figure P33.60** shows a ray that refracts into a drop at point A, is reflected from the back surface of the drop at point B, and refracts back into the air at point C. The angles of incidence and refraction, θ_a and θ_b, are shown at points A and C, and the angles of incidence and reflection, θ_a and θ_r, are shown at point B. (a) Show that $\theta_a^B = \theta_b^A$, $\theta_a^C = \theta_b^A$, and $\theta_b^C = \theta_a^A$. (b) Show that the angle in radians between the ray before it enters the drop at A and after it exits at C (the total angular deflection of the ray) is $\Delta = 2\theta_a^A - 4\theta_b^A + \pi$. (*Hint:* Find the angular deflections that occur at A, B, and C, and add them to get Δ.) (c) Use Snell's law to write Δ in terms of θ_a^A and n, the refractive index of the water in the drop. (d) A rainbow will form when the angular deflection Δ is *stationary* in the incident angle θ_a^A—that is, when $d\Delta/d\theta_a^A = 0$. If this condition is satisfied, all the rays with incident angles close to θ_a^A will be sent back in the same direction, producing a bright zone in the sky. Let θ_1 be the value of θ_a^A for which this occurs. Show that $\cos^2\theta_1 = \frac{1}{3}(n^2 - 1)$. [*Hint:* You may find the derivative formula $d(\arcsin u(x))/dx = (1 - u^2)^{-1/2}(du/dx)$ helpful.] (e) The index of refraction in water is 1.342 for violet light and 1.330 for red light. Use the results of parts (c) and (d) to find θ_1 and Δ for violet and red light. Do your results agree with the angles shown in Fig. 33.19d? When you view the rainbow, which color, red or violet, is higher above the horizon?

Figure **P33.60**



33.61 ••• **CALC** A *secondary rainbow* is formed when the incident light undergoes two internal reflections in a spherical drop of water as shown in Fig. 33.19e. (See Challenge Problem 33.60.) (a) In terms of the incident angle θ_a^A and the refractive index n of the drop, what is the angular deflection Δ of the ray? That is, what is the angle between the ray before it enters the drop and after it exits? (b) What is the incident angle θ_2 for which the derivative of Δ with respect to the incident angle θ_a^A is zero? (c) The indexes of refraction for red and violet light in water are given in part (e) of Challenge Problem 33.60. Use the results of parts (a) and (b) to find θ_2 and Δ for violet and red light. Do your results agree with the angles shown in Fig. 33.19e? When you view a secondary rainbow, is red or violet higher above the horizon? Explain.

MCAT-STYLE PASSAGE PROBLEMS

BIO Seeing Polarized Light. Some insect eyes have two types of cells that are sensitive to the plane of polarization of light. In a simple model, one cell type (type H) is sensitive to horizontally polarized light only, and the other cell type (type V) is sensitive to vertically polarized light only. To study the responses of these cells, researchers fix the insect in a normal, upright position so that one eye is illuminated by a light source. Then several experiments are carried out.

33.62 First, light with a plane of polarization at 45° to the horizontal shines on the insect. Which statement is true about the two types of cells? (a) Both types detect this light. (b) Neither type detects this light. (c) Only type H detects the light. (d) Only type V detects the light.

33.63 Next, unpolarized light is reflected off a smooth horizontal piece of glass, and the reflected light shines on the insect. Which statement is true about the two types of cells? (a) When the light is directly above the glass, only type V detects the reflected light. (b) When the light is directly above the glass, only type H detects the reflected light. (c) When the light is about 35° above the horizontal, type V responds much more strongly than type H does. (d) When the light is about 35° above the horizontal, type H responds much more strongly than type V does.

33.64 To vary the angle as well as the intensity of polarized light, ordinary unpolarized light is passed through one polarizer with its transmission axis vertical, and then a second polarizer is placed between the first polarizer and the insect. When the light leaving the second polarizer has half the intensity of the original unpolarized light, which statement is true about the two types of cells? (a) Only type H detects this light. (b) Only type V detects this light. (c) Both types detect this light, but type H detects more light. (d) Both types detect this light, but type V detects more light.

ANSWERS

Chapter Opening Question ?

(iv) The brilliance and color of a diamond are due to total internal reflection from its surfaces (Section 33.3) and to dispersion, which spreads this light into a spectrum (Section 33.4).

Key Example VARIATION Problems

VP33.2.1 (a) 70.0° (b) 31.5°
VP33.2.2 (a) 1.81 (b) 1.66×10^8 m/s
VP33.2.3 (a) 1.25 (b) 27.3° (c) 4.72×10^{14} Hz (d) 4.72×10^{14} Hz
VP33.2.4 (a) 73.5° (b) 1.04
VP33.5.1 238 W/m^2
VP33.5.2 (a) 27.0 W/m^2 (b) 33.0°
VP33.5.3 (a) 49.3 W/m^2 (b) 40.5 W/m^2 (c) 24.8 W/m^2
VP33.5.4 $5I_0/4$
VP33.6.1 (a) 60.0° (b) partially polarized
VP33.6.2 (a) 1.54 (b) 33.0°
VP33.6.3 (a) no (b) yes, 58.9°
VP33.6.4 (a) 36.5° (b) 1.80

Bridging Problem

1.93×10^8 m/s